Chemical Equilibrium and Kc

Kc expressions, Q versus K and ICE tables

Name _____ Class _____ Date _____

USEFUL RELATIONSHIPS

$a\,\mathrm{A} + b\,\mathrm{B} \rightleftharpoons c\,\mathrm{C} + d\,\mathrm{D}$: $K_\mathrm{c} = \dfrac{[\mathrm{C}]^c[\mathrm{D}]^d}{[\mathrm{A}]^a[\mathrm{B}]^b}$ (each concentration divided by $c^\circ = 1\ \mathrm{M}$)

Leave out pure solids and liquids · Reverse: $1/K$ · Multiply coefficients by n: K^n

$Q < K$: forward · $Q > K$: reverse · $Q = K$: at equilibrium

Write concentrations with units (M) and give Kc to three significant figures.

A. Understanding

1. Explain what is meant by a dynamic equilibrium. Are the concentrations of reactants and products equal at equilibrium?

B. Writing Kc

2. Write the K_c expression for: (a) $\mathrm{N_2\,(g) + 3\,H_2\,(g) \rightleftharpoons 2\,NH_3\,(g)}$ (b) $\mathrm{2\,SO_2\,(g) + O_2\,(g) \rightleftharpoons 2\,SO_3\,(g)}$ (c) $\mathrm{CaCO_3\,(s) \rightleftharpoons CaO\,(s) + CO_2\,(g)}$ (d) $\mathrm{PCl_5\,(g) \rightleftharpoons PCl_3\,(g) + Cl_2\,(g)}$

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C. Calculating Kc

3. An equilibrium mixture of $2\,SO_2\,(g) + O_2\,(g) \rightleftharpoons 2\,SO_3\,(g)$ contains 0.0400 M SO_2, 0.0200 M O_2 and 0.160 M SO_3. Calculate K_c.

4. Using your answer to question 3, find K_c for (a) $2\,SO_3 \rightleftharpoons 2\,SO_2 + O_2$ and (b) $SO_2 + \frac{1}{2}\,O_2 \rightleftharpoons SO_3$.

D. Q versus K

5. For $N_2O_4\,(g) \rightleftharpoons 2\,NO_2\,(g)$, $K_c = 0.0200$. A mixture contains 0.100 M N_2O_4 and 0.0200 M NO_2. Calculate Q and predict the direction of change.

E. ICE tables

6. 0.200 M H_2 and 0.200 M I_2 are mixed: $H_2\,(g) + I_2\,(g) \rightleftharpoons 2\,HI\,(g)$, $K_c = 49.0$. Find the equilibrium concentrations of all three gases.

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7. 0.500 M $\mathrm{PCl_5}$ is placed in a container: $\mathrm{PCl_5\,(g) \rightleftharpoons PCl_3\,(g) + Cl_2\,(g)}$. At equilibrium, $[\mathrm{Cl_2}] = 0.100$ M. Calculate K_c.

F. Challenge

8. One reaction has $K = 1 \times 10^{10}$ and another $K = 1 \times 10^{-10}$. Describe the equilibrium mixture in each case.

9. Using K_c from question 7, find the equilibrium concentrations when 0.300 M $\mathrm{PCl_5}$ is placed in a container. (You will need to solve a quadratic equation.)

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